AHMAD **HATEM**

Mechatronics Engineering Student | Specializing in Robotics & AI
Perception Engineer at ASU Racing Team

Cairo, Egypt
+201026088498
ahmadhatemmahmoud@gmail.com
linkedin.com/in/ahmadhatem33
ahmadhatemm.github.io

SUMMARY

Senior Mechatronics & Automation Engineering student (dual degree, Ain Shams University and University of East London) specializing in Robotics and AI. Perception Engineer on the ASU Racing Team's autonomous car, with hands-on work in ROS 2 and Gazebo simulation, computer vision, and AI automation. Build robots and embedded systems end to end, from CAD and PCBs to PLCs and Python.

EXPERIENCE

Perception Engineer | ASU Racing Team & ARL – Shell AI Competition SEP 2025 – Present
- Engineered and optimized perception computer vision models to detect track lanes for our autonomous car.
- Developed detailed simulation environments using ROS and Gazebo to validate LiDAR and camera sensor integration.

AI Automation Engineer Trainee | Digital Egypt Pioneers Initiative JUL 2026 – Present
- Automated a complete resume-screening system, a booking system for salons, and a grade-checking system that sends reports using n8n, python and APIs.

Mechatronics Engineer Intern | Gawad Mixers JUL 2026 – AUG 2026
- Spent a day in each production sector: casting, manufacturing, lubrication, finishing of sanitizers and mixers, and CAD design.
- Designed my own mixer in SolidWorks while reviewing the company's design files.

Automotive Engineer Intern | Engovation & Egyptian Engineers Syndicate APR 2025 – JUN 2025
- Performed real-world vehicle diagnostics, analysing internal combustion engines and powertrain elements.

Data Analyst Specialist Trainee | Google & Digital Egypt Pioneers Initiative SEP 2024 – JUL 2025
- Analysed 2,000+ hotels in Cairo, comparing average nightly prices on peak and normal calendar days, their distribution, and sizes.

Powertrain Engineer Trainee | ASU Racing Team OCT 2024 – DEC 2024
- Designed and analysed motor control circuits and custom PCBs using component datasheets; implemented STM32 embedded control to interface with hardware.

PROJECTS

Graduation Project: Multi-Robot Formation Control | Simulation Team & Formation Infrastructure Lead OCT 2026 – Present
- Goal: simulate a fleet of rover robots that move together in formation, using a ROS 2 Humble and Ignition Gazebo workspace.
- Lead the Simulation Sub-Team and own the multi-robot infrastructure: integrated the team's rover model (URDF) and built a launch file that spawns four rovers, each named and controlled through its own command topic (rover1 to rover4).

6-DOF Robotic Manipulator | ROS 2, MoveIt 2, micro-ROS OCT 2025 – JAN 2026
- Designed, simulated, and built a 4-DOF arm with a 2-DOF gripper, bridging a digital-twin simulation and real hardware.

Ecobee: AI Garbage Collector Robot | Design and embedding AI detection FEB 2026 – JUN 2026
- Built a garbage-collecting robot with AI-based object detection running on a Raspberry Pi, capturing wastes like cans and tissues using real-time YOLOv13 model connected to the Raspberry Pi online.

3-DOF Delta Robot | Advanced Robotics course MAR 2026 – MAY 2026
- Designed, manufactured, and controlled a delta robot driven by two stepper motors and one DC motor.

Aluminum Air-Powered Car | Mechanical design & manufacturing MAY 2026 – JUL 2026
- Designed, manufactured, and tested an aluminum vehicle propelled by compressed air from a 3-litre, 10-bar tank, integrating fluid, mechanical, and electrical subsystems with braking and control.

Buck Converter & 5V Power Supply | Power electronics APR 2025 – JUN 2025
- Designed and built a PWM buck converter for a DC motor at variable voltage, and a regulated 5 V DC supply from 220 V AC mains.

Smart Production System & PID Water Tank Level | Siemens TIA Portal MAY 2026 – JUN 2026
- Programmed and simulated an automated manufacturing line (Factory I/O, PLCSim) and a liquid-level controller on an S7-1200 PLC.

EDUCATION

Mechatronics and Automation Engineering | Ain Shams University & University of East London SEP 2023 – JUN 2027

A Levels | Kaumeya International British School SEP 2020 – JUN 2023

CERTIFICATIONS

- IBM Machine Learning with Python (2025)
- IBM AI Fundamentals (2025)
- Microsoft Azure AI Fundamentals (2025)
- MATLAB Simulink Onramp and Simscape Onramp (2025)

TECHNICAL SKILLS

- Programming: Python, C/C++, JavaScript, HTML, CSS, Java, MATLAB / Simulink, Stateflow
- Robotics & simulation: ROS 2 (Humble), Gazebo, MoveIt 2, RViz2, micro-ROS, URDF/Xacro, LiDAR and camera integration
- Automation & controls: PID control, Siemens TIA Portal (S7-1200 PLC), PLCSim, Factory I/O
- Embedded & CAD: ESP32, Arduino, STM32, Raspberry Pi; I2C, SPI, UART; SolidWorks, Autodesk Inventor